

## CL117-INTRODUCTION TO INFORMATION AND COMMUNICATION TECHNOLOGIES

### What exactly is VLOOKUP?

Basically, VLOOKUP lets you **search for specific information** in your spreadsheet. For example, if you have a list of products with prices, you could search for the price of a specific item.

We're going to use VLOOKUP to find the price of the **Photo frame**. You can probably already see that the price is \$9.99, but that's because this is a simple example. Once you learn how to use VLOOKUP, you'll be able to use it with larger, more complex spreadsheets, and that's when it will become truly useful.

|    | A                  | B       | C | D | E | F |
|----|--------------------|---------|---|---|---|---|
| 1  | Item               | Price   |   |   |   |   |
| 2  | Spice rack         | \$19.99 |   |   |   |   |
| 3  | Stationery         | \$5.49  |   |   |   |   |
| 4  | Gift basket        | \$25.99 |   |   |   |   |
| 5  | Cutting board      | \$24.99 |   |   |   |   |
| 6  | Landscape painting | \$35.99 |   |   |   |   |
| 7  | Greeting card      | \$4.99  |   |   |   |   |
| 8  | T-shirt            | \$15.49 |   |   |   |   |
| 9  | Scarf              | \$29.99 |   |   |   |   |
| 10 | Coffee mug         | \$8.99  |   |   |   |   |
| 11 | Tea set            | \$16.99 |   |   |   |   |
| 12 | Serving bowl       | \$12.99 |   |   |   |   |
| 13 | Wrapping paper     | \$3.99  |   |   |   |   |
| 14 | Photo frame        | \$9.99  |   |   |   |   |
| 15 | Handmade soap      | \$4.49  |   |   |   |   |
| 16 | Gourmet hot cocoa  | \$5.99  |   |   |   |   |

We'll add our formula to cell **E2**, but you can add it to any blank cell. As with any formula, you'll start with an equals sign (=). Then type the formula name. Our arguments will need to be in parentheses, so type an open parenthesis. So far, it should look like this:

**=VLOOKUP(**

### *Adding the arguments*

Now, we'll add our arguments. The arguments will tell VLOOKUP what to search for and where to search.

The first argument is the name of the item you're searching for, which in this case is Photo frame. Because the argument is text, we'll need to put it in double quotes:

**=VLOOKUP("Photo frame"**

The second argument is the **cell range that contains the data**. In this example, our data is in A2:B16. As with any function, you'll need to use a comma to separate each argument:

**=VLOOKUP("Photo frame", A2:B16**

Note: It's important to know that VLOOKUP will always search the first column in this range. In this example, it will search column A for "Photo frame". In some cases, you may need to move the columns around, so the first column contains the correct data.

The third argument is the column index number. It's simpler than it sounds: The first column in the range is 1, the second column is 2, etc. In this case, we are trying to find the price of the item, and the prices are contained in the second column. This means our third argument will be 2:

**=VLOOKUP("Photo frame", A2:B16, 2**

The fourth argument tells VLOOKUP whether to look for **approximate matches**, and it can be either TRUE or FALSE. If it is TRUE, it will look for approximate matches. Generally, this is only useful if the first column has numerical values that have been sorted. Because we're only looking for exact matches, the fourth argument should be **FALSE**. This is our last argument, so go ahead and close the parentheses:

**=VLOOKUP("Photo frame", A2:B16, 2, FALSE)**

That's it! When you press Enter, it should give you the answer, which is **9.99**.

|                                              |   |      |   |   |  |
|----------------------------------------------|---|------|---|---|--|
| fx =VLOOKUP("Photo frame", A2:B16, 2, FALSE) |   |      |   |   |  |
| C                                            | D | E    | F | G |  |
|                                              |   |      |   |   |  |
|                                              |   | 9.99 |   |   |  |
|                                              |   |      |   |   |  |

### How it works

Let's take a look at how this formula works. It first **searches vertically** down the first column (VLOOKUP is short for **vertical lookup**). When it finds "Photo frame", it **moves to the second column** to find the price.



|    | A                 | B       | C | D |
|----|-------------------|---------|---|---|
| 1  | Item              | Price   |   |   |
| 2  |                   | \$9.99  |   |   |
| 3  |                   | \$9.99  |   |   |
| 4  |                   | \$9.99  |   |   |
| 5  |                   | \$9.99  |   |   |
| 6  |                   | \$9.99  |   |   |
| 7  | Greeting card     | \$4.99  |   |   |
| 8  | T-shirt           | \$15.49 |   |   |
| 9  | Scarf             |         |   |   |
| 10 | Coffee mug        |         |   |   |
| 11 | Tea set           |         |   |   |
| 12 | Serving bowl      |         |   |   |
| 13 | Wrapping paper    |         |   |   |
| 14 | Photo frame       | \$9.99  |   |   |
| 15 | Handmade soap     | \$4.49  |   |   |
| 16 | Gourmet hot cocoa | \$5.99  |   |   |
| 17 |                   |         |   |   |

First, it searches  
the first column for  
"Photo frame"

Then, it finds  
the price in the  
second column

If we want to find the price of a different item, we can just change the first argument:

**=VLOOKUP("T-shirt", A2:B16, 2, FALSE)**

or:

**=VLOOKUP("Gift basket", A2:B16, 2, FALSE)**

Another example

Are you ready for a slightly more complicated example? Let's say we have a third column that has the category for each item. This time, instead of finding the price we'll find the category.

|    | A                  | B       | C        |
|----|--------------------|---------|----------|
| 1  | Item               | Price   | Category |
| 2  | Spice rack         | \$19.99 | Kitchen  |
| 3  | Stationery         | \$5.49  | Writing  |
| 4  | Gift basket        | \$25.99 | Gifts    |
| 5  | Cutting board      | \$24.99 | Kitchen  |
| 6  | Landscape painting | \$35.99 | Art      |
| 7  | Greeting card      | \$4.99  | Gifts    |
| 8  | T-shirt            | \$15.49 | Clothing |
| 9  | Scarf              | \$29.99 | Clothing |
| 10 | Coffee mug         | \$8.99  | Kitchen  |
| 11 | Tea set            | \$16.99 | Kitchen  |
| 12 | Serving bowl       | \$12.99 | Kitchen  |
| 13 | Wrapping paper     | \$3.99  | Gifts    |
| 14 | Photo frame        | \$9.99  | Gifts    |
| 15 | Handmade soap      | \$4.49  | Gifts    |
| 16 | Gourmet hot cocoa  | \$5.99  | Food     |



To find the category, we'll need to **change the second and third arguments** in our formula. First, we'll change the range to **A2:C16** so it includes the third column. Next, we'll change the column index number to **3** because our categories are in the third column:

**=VLOOKUP("Gift basket", A2:C16, 3, FALSE)**

When you press Enter, you'll see that the Gift basket is in the **Gifts** category.

|    |       |                                           |   |   |  |  |
|----|-------|-------------------------------------------|---|---|--|--|
| fx |       | =VLOOKUP("Gift basket", A2:C16, 3, FALSE) |   |   |  |  |
| D  | E     | F                                         | G | H |  |  |
|    | Gifts |                                           |   |   |  |  |

**Note:** If the [\[range\\_lookup\]](#) argument in the Vlookup function is set to TRUE (or is omitted), this tells the Vlookup function that if it is unable to find an exact match to the [lookup\\_value](#), then the closest match below the [lookup\\_value](#) should be used instead.

## Data Visualization

In previous lab we have used charts in excel to make a graphical representation of any set of data. Now we can do extensive data analysis using PivotTables and produce desired reports. The integration of Data Model with PivotTable enhances the way the data is collated, connected, summarized and reported. You can import tables from external data sources and create a PivotTable with the imported tables. This facilitates automatic updations of the values in the PivotTable whenever the data in the

Now let's see How can I build a Pivot Table in MS Excel.

Pivot Table is one of the most powerful and useful tool in MS Excel.

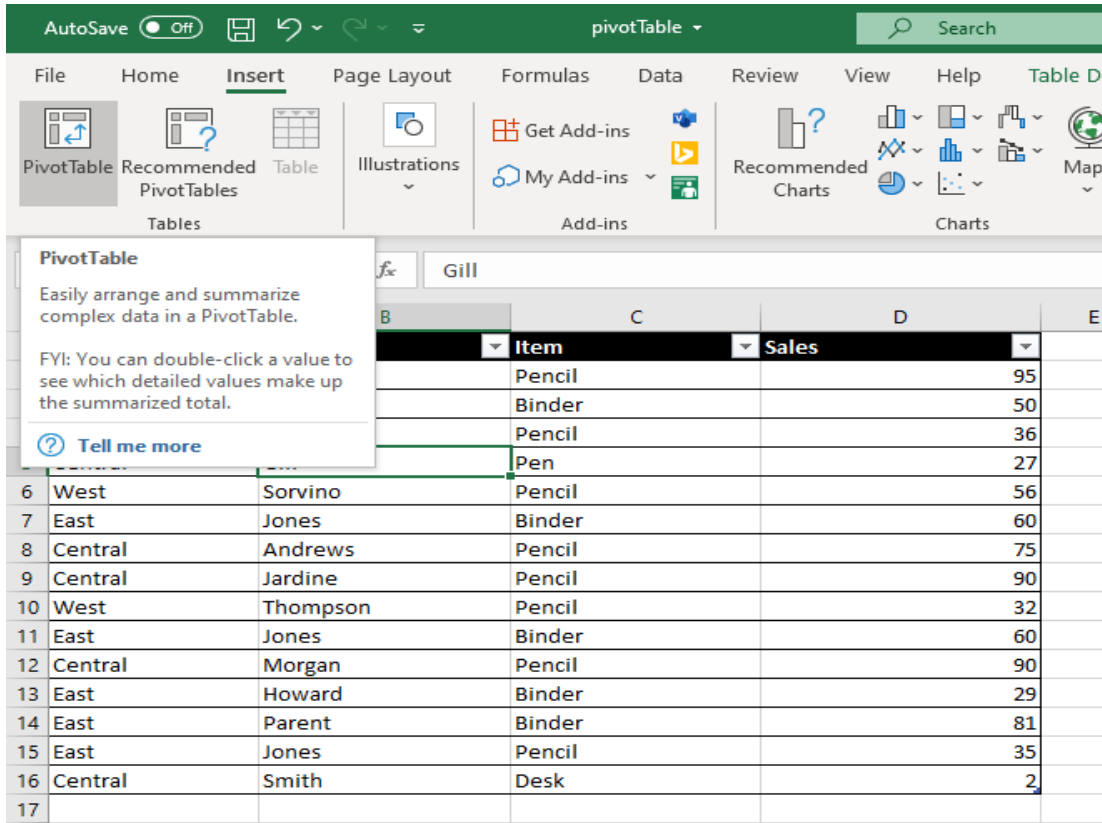
## Sample Data

Create the following Data in an Excel Worksheet

|    | A       | B        | C      | D     | E |
|----|---------|----------|--------|-------|---|
| 1  | Region  | Rep      | Item   | Sales |   |
| 2  | East    | Jones    | Pencil | 95    |   |
| 3  | Central | Kivell   | Binder | 50    |   |
| 4  | Central | Jardine  | Pencil | 36    |   |
| 5  | Central | Gill     | Pen    | 27    |   |
| 6  | West    | Sorvino  | Pencil | 56    |   |
| 7  | East    | Jones    | Binder | 60    |   |
| 8  | Central | Andrews  | Pencil | 75    |   |
| 9  | Central | Jardine  | Pencil | 90    |   |
| 10 | West    | Thompson | Pencil | 32    |   |
| 11 | East    | Jones    | Binder | 60    |   |
| 12 | Central | Morgan   | Pencil | 90    |   |
| 13 | East    | Howard   | Binder | 29    |   |
| 14 | East    | Parent   | Binder | 81    |   |
| 15 | East    | Jones    | Pencil | 35    |   |
| 16 | Central | Smith    | Desk   | 2     |   |
| 17 |         |          |        |       |   |
| 18 |         |          |        |       |   |

## Inserting a pivot Table

1. To start off, select any cell in the data and click Pivot Table on the Insert tab of the ribbon:



**PivotTable**

Easily arrange and summarize complex data in a PivotTable.

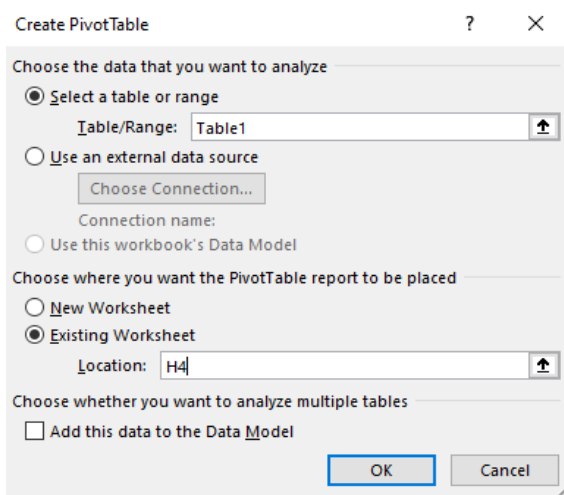
FYI: You can double-click a value to see which detailed values make up the summarized total.

[Tell me more](#)

|    | Item    | Sales    |        |    |
|----|---------|----------|--------|----|
|    | Pencil  | 95       |        |    |
|    | Binder  | 50       |        |    |
|    | Pencil  | 36       |        |    |
|    | Pen     | 27       |        |    |
| 6  | West    | Sorvino  | Pencil | 56 |
| 7  | East    | Jones    | Binder | 60 |
| 8  | Central | Andrews  | Pencil | 75 |
| 9  | Central | Jardine  | Pencil | 90 |
| 10 | West    | Thompson | Pencil | 32 |
| 11 | East    | Jones    | Binder | 60 |
| 12 | Central | Morgan   | Pencil | 90 |
| 13 | East    | Howard   | Binder | 29 |
| 14 | East    | Parent   | Binder | 81 |
| 15 | East    | Jones    | Pencil | 35 |
| 16 | Central | Smith    | Desk   | 2  |
| 17 |         |          |        |    |

Excel will display the Create Pivot Table window. Notice the data range is already filled in. The default location for a new pivot table is New Worksheet.

- Override the default location and enter H4 to place the pivot table on the current worksheet:



Create PivotTable

Choose the data that you want to analyze

☒ Select a table or range

Table/Range: Table1

☐ Use an external data source

Choose Connection...

Connection name:

☐ Use this workbook's Data Model

Choose where you want the PivotTable report to be placed

☐ New Worksheet

☒ Existing Worksheet

Location: H4

Choose whether you want to analyze multiple tables

☐ Add this data to the Data Model

OK Cancel

- Click OK, and Excel builds an empty pivot table starting in cell H4.

|    | A       | B        | C      | D     | E | F | G | H | I | J | K | L |
|----|---------|----------|--------|-------|---|---|---|---|---|---|---|---|
| 1  | Region  | Rep      | Item   | Sales |   |   |   |   |   |   |   |   |
| 2  | East    | Jones    | Pencil | 95    |   |   |   |   |   |   |   |   |
| 3  | Central | Kovell   | Binder | 50    |   |   |   |   |   |   |   |   |
| 4  | Central | Jardine  | Pencil | 36    |   |   |   |   |   |   |   |   |
| 5  | Central | Gill     | Pen    | 27    |   |   |   |   |   |   |   |   |
| 6  | West    | Sorvino  | Pencil | 56    |   |   |   |   |   |   |   |   |
| 7  | East    | Jones    | Binder | 60    |   |   |   |   |   |   |   |   |
| 8  | Central | Andrews  | Pencil | 75    |   |   |   |   |   |   |   |   |
| 9  | Central | Jardine  | Pencil | 90    |   |   |   |   |   |   |   |   |
| 10 | West    | Thompson | Pencil | 52    |   |   |   |   |   |   |   |   |
| 11 | East    | Jones    | Binder | 60    |   |   |   |   |   |   |   |   |
| 12 | Central | Morgan   | Pencil | 90    |   |   |   |   |   |   |   |   |
| 13 | East    | Howard   | Binder | 29    |   |   |   |   |   |   |   |   |
| 14 | East    | Parent   | Binder | 81    |   |   |   |   |   |   |   |   |
| 15 | East    | Jones    | Pencil | 35    |   |   |   |   |   |   |   |   |
| 16 | Central | Smith    | Desk   | 2     |   |   |   |   |   |   |   |   |
| 17 |         |          |        |       |   |   |   |   |   |   |   |   |
| 18 |         |          |        |       |   |   |   |   |   |   |   |   |
| 19 |         |          |        |       |   |   |   |   |   |   |   |   |
| 20 |         |          |        |       |   |   |   |   |   |   |   |   |
| 21 |         |          |        |       |   |   |   |   |   |   |   |   |
| 22 |         |          |        |       |   |   |   |   |   |   |   |   |

Excel also displays the PivotTable Fields pane, which is empty at this point. Note all four fields are listed, but unused:

PivotTable Fields

Choose fields to add to report:

☐ Region  
☐ Rep  
☐ Item  
☐ Sales  
 More Tables...

Drag fields between areas below:

Filters

Columns

Rows

Values

☐ Defer Layout Update
 

Update

To build a pivot table, drag fields into one the Columns, Rows, or Values area. The Filters area is used to apply global filters to a pivot table.

Let's assume we want to see **Region wise sales for each Item**, and We want to filter the Data based on the Person (Rep) who sells it.

In Order to create this Table, we have to do the following changes in the Pivot Table field.

### PivotTable Fields

Choose fields to add to report:
⚙️

☒ Region

☒ Rep

☒ Item

☒ Sales

More Tables...

---

Drag fields between areas below:

**Filters**

Rep

**Columns**

Item

**Rows**

Region

**Values**

Sum of Sales

☐ Defer Layout Update
Update

Note that we wanted to see region wise total sales of each item, so we have added regions and items as Rows and Columns and Sales as Values. And we wanted to filter the data based on the Receptionist who sells it. This will Produce the Following Result.

|              |        |      |     |        |             |     |
|--------------|--------|------|-----|--------|-------------|-----|
| Rep          | (All)  |      |     |        |             |     |
| Sum of Sales | Items  |      |     |        |             |     |
| Region       | Binder | Desk | Pen | Pencil | Grand Total |     |
| Central      |        | 50   | 2   | 27     | 291         | 370 |
| East         |        | 230  |     |        | 130         | 360 |
| West         |        |      |     |        | 88          | 88  |
| Grand Total  |        | 280  | 2   | 27     | 509         | 818 |

Lets test our Pivot table now and apply filter to find total sales done by the Rep named **Gill**.

Go to the filter and Expand to see all the values of Rep Column



|                                                |        |
|------------------------------------------------|--------|
| Rep                                            | (All)  |
| Search                                         |        |
| (All)                                          |        |
| Andrews                                        |        |
| <b>Gill</b>                                    |        |
| Howard                                         |        |
| Jardine                                        |        |
| Jones                                          |        |
| Kivell                                         |        |
| Morgan                                         |        |
| Parent                                         |        |
| <input type="checkbox"/> Select Multiple Items |        |
| OK                                             | Cancel |

| Desk | Pen | Pencil | Grand Total |
|------|-----|--------|-------------|
| 2    | 27  | 291    | 370         |
|      |     | 130    | 360         |
|      |     | 88     | 88          |
| 2    | 27  | 509    | 818         |

Select the name Gill from Dropdown menu and following result will be displayed showing the Region wise sales of each item where Rep was

|              |       |
|--------------|-------|
| Rep          | Gill  |
| Sum of Sales | Items |
| Region       | Pen   |
| Central      | 27    |
| Grand Total  | 27    |

### Task 1:

Now lets assume we want to see the average sales instead of total sales. All you have to do is change the Sum to Average option.

The output of the Pivot table should be like this.

|                        |        |      |     |        |             |  |  |
|------------------------|--------|------|-----|--------|-------------|--|--|
| Rep                    | (All)  |      |     |        |             |  |  |
| Average of Sales Items |        |      |     |        |             |  |  |
| Region                 | Binder | Desk | Pen | Pencil | Grand Total |  |  |
| Central                | 50     | 2    | 27  | 72.75  | 52.85714286 |  |  |
| East                   | 57.5   |      |     | 65     | 60          |  |  |
| West                   |        |      |     | 44     | 44          |  |  |
| Grand Total            | 56     | 2    | 27  | 63.625 | 54.53333333 |  |  |